

Semantic Reflex Layer

Fast semantic control between perception and physics

SYSTEM 1 CONCEPT

Core idea

A Semantic Reflex Layer is a fast, bounded decision layer between structured perception and low-level control. Ollaya scores known semantic actions with confidence; it does not directly control complex physics.

Control hierarchy

- WHY - System 2: goals, planning, novel reasoning and policy evolution.
- WHAT - Semantic Reflex Layer: select the appropriate semantic behaviour now.
- HOW - Planner/controller: trajectory, path, speed profile and execution logic.
- DO - Low-level control: PID/MPC, steering, throttle, brake and actuators.

Two-loop operating model

- Fast loop: structured state -> Ollaya -> semantic decision -> controller -> environment -> feedback.
- Slow loop: low confidence, novelty or repeated failure escalates to a large private LLM.
- System 2 may propose policy/YAML changes; validate, replay-test and version before deployment.

Why it matters

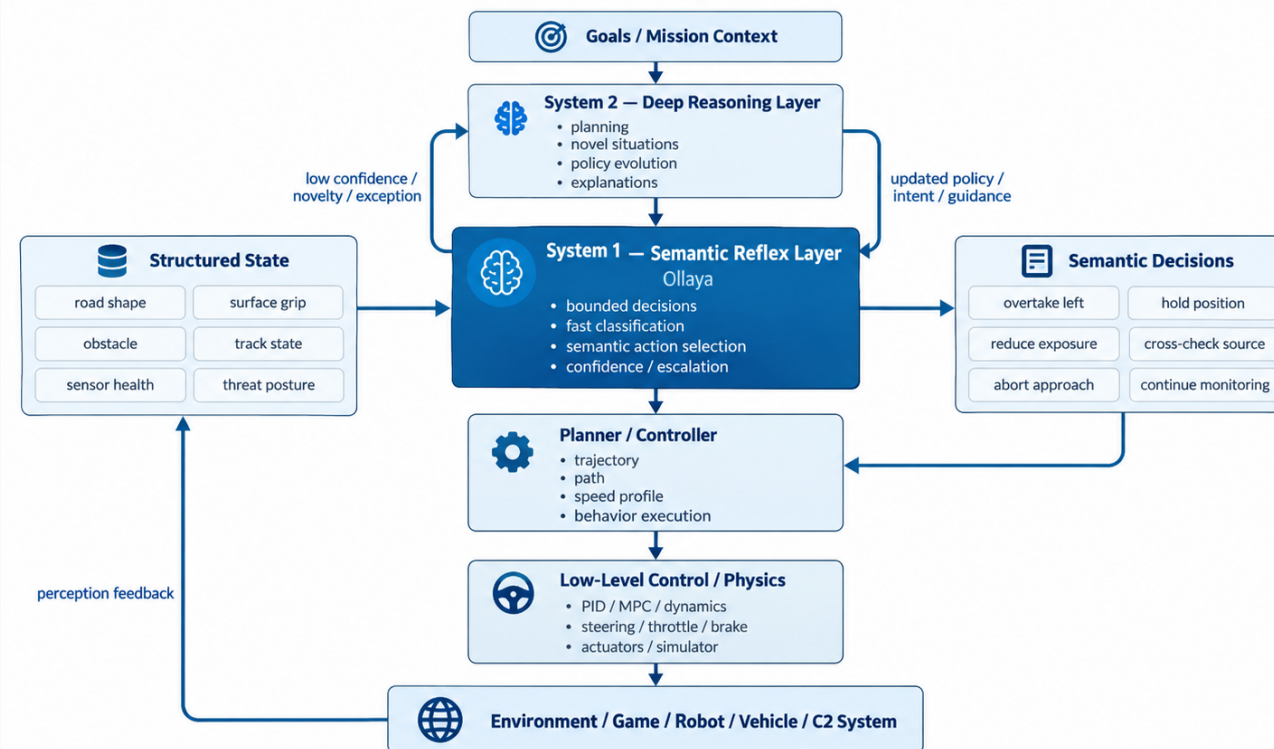
- Low steady-state compute: no generative LLM is required in the permanent fast loop.
- Explainable and portable: trace state + probabilities + policy version across C2, games, robots and vehicles.

Design principle

Use System 1 for fast semantic reflexes; use System 2 only when the situation exceeds the known policy.

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Control Hierarchy

1 WHY → System 2

2 WHAT → Semantic Reflex Layer

3 HOW → Planner / Controller

4 DO → Low-Level Control / Physics

Ollaya provides semantic control, not direct physics control.